

Liquidity and Autocorrelations in Individual Stock Returns

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Abstract

This paper documents a strong relationship between short-run reversals and stock return illiquidity, even after controlling for trading volume. The largest reversals and the potential contrarian trading strategy profits occur in the high turnover, low liquidity stocks, as the price pressures caused by non-informational demands for immediacy are accommodated. Thus, the high frequency negative autocorrelations are more likely to result from stresses in the market for liquidity. The contrarian trading strategy profits are smaller than the likely transactions costs because the high turnover, low liquidity stocks face large transaction and market impact costs. This lack of profitability and the fact that the overall findings are consistent with rational equilibrium paradigms suggest that the violation of the efficient market hypothesis due to short-term reversals is not so egregious after all.

JEL Classifications: G11, G12, C11

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1 Introduction

This paper examines two important paradigms in finance: (i) the price impact of market making costs; and (ii) the efficient market hypothesis. Consider first the price impact of market making costs. The market microstructure literature has studied the incorporation of information into prices through the trading process. Of course, the trading process is subject to frictions that impact asset prices.¹ Such frictions may arise due to informational asymmetries or due to the inventory holding costs of market makers.² In this paper, we consider a model motivated by inventory holding costs of market making to study short-run reversals in individual stock returns. Turning now to the second important paradigm of efficient markets, the prior documented reversals have challenged the concept of market efficiency even in its weakest form per the seminal review of Fama (1970). In particular, Lo and MacKinlay (1988, 1990) find that weekly portfolio returns have large positive autocorrelations, which they ascribe to the cross-autocorrelation patterns in stock returns. Moreover, Lehmann (1990) and Jegadeesh (1990) show that contrarian strategies that exploit the return reversals in individual stocks generate abnormal returns of about 1.7% per week and 2.5% per month, respectively.

From a theoretical point of view, the potentially high abnormal payoffs to short-run contrarian strategies have been explained by Campbell, Grossman, and Wang (1993) (henceforth CGW) within a rational equilibrium framework. The CGW model does not allow for frictions based on information asymmetries and is analogous to the inventory cost models of market making, noted earlier. The basic idea is that demand shocks from non-informational traders is absorbed by market makers who require compensation for holding sub-optimal inventories. Specifically, non-informational trading causes price movements that, when absorbed by liquidity suppliers, cause prices to revert. Such non-informed trading is accompanied with high trading volume, whereas informed trading is accompanied with little trading volume. Thus, price changes accompanied with high (low) trading volume should (should not) revert. Empirical support for the role of trading volume advocated by CGW has been provided by Conrad, Hameed, and Niden (1994) (henceforth CHN). Using a sample of NASDAQ stocks, CHN find that reversal profitability increases with trading activity. Cooper (1999), on the other hand, uses a sample of large NYSE-AMEX stocks and finds evidence to the contrary. Cooper's findings provide support for the asymmetric information model of Wang (1994) in which price continuations are accompanied with high trading volumes when informed investors

¹See, e.g., Stoll (2000).

²See Stoll (1978) and Ho and Stoll (1981, 1983) for inventory based market making models.

condition their trades on private information.

The CGW model motivates our work. In particular, it implicitly assumes that the demand curves for stocks are not perfectly elastic; otherwise, there would be no impact of trades on prices and subsequent price reversals would not occur. With downward sloping demand curves, price reversals should follow non-informational liquidity trading. In particular, non-informational demands for liquidity cause price pressures that are subsequently reversed as liquidity suppliers react to potential profit opportunities attributable to prices that move away from fundamentals. Thus, the CGW model points to a potential role for liquidity that need not be captured by trading volume based measures. Consequently, this paper incorporates the unexplored dimension of liquidity into the empirical investigation of the serial correlation patterns of individual stock returns and the overall profitability of short-run contrarian strategies.

We conjecture that the price pressures caused by demands for immediacy from liquidity traders lead to, or at least enhance, price reversals. Therefore, controlling for trading volume, liquidity, or rather the lack of liquidity, should have an impact on stock return autocorrelations. While liquidity is an elusive concept, most market participants agree that liquidity generally reflects the ability to buy or sell sufficient quantities, quickly, at low trading cost and without impacting the market price too much. Following Amihud (2002), we measure weekly as well as monthly illiquidity as the average of the daily price impacts of the order flow, i.e., the daily absolute price change per dollar of daily trading volume. The Amihud measure has been inspired by the Kyle (1985) λ , the market impact measure. It should be noted that illiquidity and trading volume are markedly different both conceptually as well as based on data - they measure different things and are only mildly correlated. Thus, illiquidity can potentially contribute to our understanding of short-term return reversals, and, as we demonstrate, it does.

Specifically, studying a sample of NYSE-AMEX stocks over the 1962-2002 period, we find that there are reversals in weekly and monthly stock returns but they are mainly confined to the loser stocks – stocks with negative returns in the past week or month. At the weekly frequency, high turnover stocks exhibit higher negative serial correlations than low turnover stocks. The empirical evidence thus supports the predictions of CGW and is consistent with the CHN findings. The fact that our results are apparently at odds with the predictions of Wang (1994) suggests that the degree of information asymmetry in our sample of NYSE-AMEX stocks may not be large enough to have an impact on individual stock return autocorrelations. Indeed, when the degree of asymmetric information is low, the equilibrium framework of Wang reduces to that of CGW.

Next, we show that liquidity has a robust effect on stock return autocorrelations, even after controlling for trading volume, as conjectured. In particular, there is substantially more reversal in less liquid stocks than in the highly liquid stocks. Moreover, we demonstrate that the high turnover, low liquidity stocks have the highest negative serial correlations in cross-sectional regressions, and they have the highest potential contrarian strategy trading profits. The high-turnover low-liquidity stocks exhibit the largest initial price changes that are subsequently reversed as the demand for immediacy is accommodated by liquidity suppliers. That is, these high turnover, low liquidity stocks reverse the most *both* because they have the largest initial price change *and* because they have the most negative return autocorrelations.

We find the impact of liquidity on autocorrelations to be similar at the weekly and monthly frequencies. In contrast, at the monthly frequency the impact of turnover on autocorrelations reverses relative to that based on the weekly frequency; low turnover stocks exhibit more reversals than high turnover stocks. This suggests that turnover may be a poor proxy for non-informational trades at the monthly frequency. Overall, our findings show that non-informational demands for liquidity cause price pressures that are subsequently reversed as liquidity suppliers react to potential profit opportunities that are attributable to prices that move away from fundamentals. Moreover, our findings are consistent with the notion that the predictability in short horizon returns occurs because of stresses in the market for liquidity.

Indeed, Lehmann (1990) and Jegadeesh (1990) suggest that contrarian trading strategy profits are as large as 1.7% per week and 2.5% per month, respectively. Presumably, such large profits may reflect a violation of weak-form market efficiency. However, we demonstrate that a high frequency trading strategy that attempts to profit from the negative serial correlations generates high transactions costs and substantial price impact. Specifically, we use the transaction cost estimates of Keim and Madhavan (1997) within a dynamic framework to estimate net returns based on a strategy that buys the losers and sells the winners. We also consider the market impact cost analysis of Korajczyk and Sadka (2004). The evidence conclusively shows that potential profits are overwhelmed by transactions costs under each and every choice of liquidity and turnover. Consequently, while the presence of statistically significant negative autocorrelations in individual security returns is undeniable, it is not possible to profit from this short-run predictability. This lack of profitability is consistent with the Jensen (1978) definition of market efficiency and the Rubinstein (2001) definition of minimally rational markets.³

³The concept of market efficiency with respect to an information set has been defined by Jensen

To summarize, this paper provides two main contributions. First, we test the implicit CGW assumption of downward sloping demand curves by explicitly accounting for illiquidity in the price reversal strategies. We show that, over and above the impact of turnover, illiquidity has a consistent impact on short-run reversals both at the weekly and monthly frequencies. Second, we show that the potential contrarian strategy profits do not reflect violations of market efficiency because such profits are overwhelmed by transactions costs. The lack of profitability from short-run contrarian strategies and the fact that the overall findings are consistent with the rational equilibrium paradigm of CGW suggest that the violations of the efficient market hypothesis due to short term reversals are not so egregious after all.

The remainder of the paper proceeds as follows. Section 2 presents the data. Section 3 presents the results. A detailed discussion of transactions costs is presented in section 4 and section 5 concludes.

2 Data

We obtain the return and trading volume data from CRSP for the universe of NYSE-AMEX stocks over the sample period 1962 through 2002. The average number of stocks in the sample is 2070. The Amihud (2002) illiquidity measure is the average daily price impact of order flow and is computed as the absolute price change per dollar of daily trading volume:⁴

$$ILLIQ_{it} = \frac{1}{D_{it}} \sum_{t=1}^{D_{it}} \frac{|R_{itd}|}{DVOL_{itd}} * 10^6, \quad (1)$$

where R_{itd} is the daily return, $DVOL_{itd}$ is the dollar trading volume of stock i on day d in week (or month) t , and D_{it} is the number of days in week (or month) t for which data is available for stock i . We compute the Amihud illiquidity measure for both the weekly as well as the monthly frequency. In the weekly analysis, we require that a stock trades every day of the week before it is included in the sample for that week.⁵ In the monthly analysis, we require at least ten days with trades each month. In order to avoid

(1978) as the inability to profit from that information. Rubinstein (2001) defines this as minimally rational markets.

⁴Hasbrouck (2003) compares effective and price impact measures estimated from daily data to those from high frequency data and finds that the Amihud (2002) measure is the most highly correlated with trade-based measures.

⁵We have checked that our results are robust to requiring only two days of trading in each week.

the extremely illiquid stocks, penny stocks, i.e., stocks with prices less than one dollar, have been eliminated from the sample.⁶

The mean (median) cross-sectional return is 0.24% (−0.09%) per week and the mean (median) weekly turnover (defined as dollar trading volume divided by the market capitalization) is 0.24% (0.15%). The mean (median) cross-sectional illiquidity measure is 1.45 (0.16); i.e., the mean absolute daily return for one million dollar trading volume is 1.45. The correlation between returns and turnover is positive at 0.13. The unconditional correlation between returns and illiquidity is essentially zero. However, conditional on positive (negative) returns the correlation is 0.09 (−0.21) suggesting that an increase in illiquidity results in an increase in returns. This is consistent with the hypothesis that more illiquid stocks earn higher expected returns. Since turnover has often been used as a measure of liquidity⁷ one may expect the illiquidity measure to be related to turnover. However, the cross-sectional correlation between turnover and liquidity is only −0.08. The information in the Amihud illiquidity measure is not subsumed by that in turnover, thus, allowing us to separately study the impact of turnover as well as of illiquidity on the serial correlation pattern of weekly (and monthly) stock returns.

3 Results

Before we describe our findings in detail, we note that Lehmann (1990) and Jegadeesh (1990) document negative serial correlation in the cross-section of weekly and monthly stock returns, respectively. In particular, they show that contrarian strategies that exploit the return reversals in individual stocks generate abnormal returns of about 1.7% per week (Lehmann) and 2.5% per month (Jegadeesh). This economically significant predictability seems to be at odds with the notion of weak-form market efficiency. The notion is that prices should follow a martingale process over short time intervals since systematic short-run changes in fundamental values should be negligible in efficient markets with a random information arrival process.

In this paper, we conjecture that a lack of liquidity around large price changes causes deviations from the so-called martingale process and price reversals arise as the price pressure in response to a demand for liquidity from uninformed investors abates.

⁶In the context of long-term contrarian investment strategies, Ball, Kothari, and Shanken (1995) show that microstructure issues can create severe biases amongst low-priced stocks.

⁷See, e.g., Brennan, Chordia, and Subrahmanyam (1998) and Chordia, Subrahmanyam, and Anshuman (2001).

Our conjecture builds on CGW who argue that price reversals occur as risk averse market makers absorb order flow from uninformed or liquidity traders. In their model, a decline (increase) in stock prices could occur due to public information or due to liquidity driven selling (buying) pressure. When the price change is due to informational reasons, price reversals are unlikely. However, suppliers of liquidity who accommodate the non-informational sell orders demand higher expected returns and must be compensated for bearing portfolio risk when buying shares that they would otherwise not trade. In fact, liquidity suppliers are attracted by prices that move away from fundamentals in response to the liquidity driven buying or selling pressure. We do not expect price changes after the public information has been incorporated into prices. In contrast, price reversals are to be expected following liquidity driven trading. Of course, if demand curves were perfectly elastic, price reversals would not obtain. Given downward sloping demand curves, price reversals should follow liquidity or non-informational trading. The CGW model, therefore, suggests that liquidity or the lack of it should have an impact on price reversals. Consequently, in our empirical analysis, we form portfolios sorted on returns, trading volume, and illiquidity.

Below, we describe the empirical findings.

3.1 Illiquidity and Reversal Profitability

First, we test the hypothesis that it is the price pressure that causes the initial large returns in period $t - 1$, which are then reversed in period t . The price pressure should be the largest in the least liquid stocks and in stocks with high turnover where the demand for liquidity is the strongest.

Each week, we sort stocks based on the Wednesday to Tuesday close returns in week $t - 1$. Following Lehmann (1990) and Jegadeesh (1990), we skip a day at the end of each week in order to avoid the negative serial correlation induced by the bid-ask bounce.⁸ We further sort negative and positive return portfolios into the extreme and non-extreme portfolios. That is, we form four portfolios: the first is the extreme negative return

⁸We follow the skip-day methodology throughout the rest of the paper. In unreported results, we have checked to see the impact on the profitability of reversal strategies when we do not skip a day between the formation and holding periods. The potential profits (including the impact of the bid-ask bounce) are indeed higher. However, it should be noted that in the next section we will comprehensively study the impact of transaction costs and show that the reversal profits do not survive reasonable costs. This evidence is robust to whether or not we skip one day. All results based on full five-day conditioning past returns are available upon request.

portfolio with an average equally weighted return of -5.69% per week; the second is the medium negative return portfolio; the third is the medium positive return portfolio; and the fourth is the extreme positive return portfolio with an average return of 6.11% per week. We form a total of 64 portfolios by sorting independently on illiquidity and turnover within each return portfolio. The turnover and illiquidity portfolio one (four) has the lowest (highest) turnover and illiquidity, respectively.⁹

Before we proceed with the analysis, we note that in untabulated results (available upon request) we find that, for the loser stocks, for any illiquidity portfolio the week $t - 1$ equally weighted average return decreases with turnover, and for any turnover portfolio the week $t - 1$ return decreases with illiquidity. For any illiquidity portfolio, the return differential between turnover portfolios four and one is negative and this difference becomes monotonically more negative as illiquidity increases. Similarly, for any turnover portfolio, the return differential between illiquidity portfolios four and one is negative and this difference becomes monotonically more negative as turnover increases. An analogous pattern obtains for the extreme winner stocks as well. These return patterns are depicted in Panel A of Figure 1. In sum, during the formation period, the impact of price pressure increases with illiquidity and turnover. The large price changes occur in the high turnover portfolios and in portfolios with high illiquidity.

Next, CGW have suggested that non-informational or liquidity trades cause high turnover. Thus, the demand for liquidity from the non-informational trades combined with illiquidity causes large price changes. We conjecture that the large price changes in week $t - 1$, noted in the previous paragraph, should be reversed in week t as liquidity suppliers step in to profit from prices that temporarily deviate from fundamentals. We now check for reversals amongst the high turnover and the high illiquidity stocks in week t .

Table 1 presents the equally weighted average returns and standard deviations for the three-way sorted portfolios in week t . Recall that return portfolio one (four) has the lowest (highest) returns in week $t - 1$. Also, turnover and illiquidity portfolio one (four) has the lowest (highest) turnover and illiquidity, respectively. Focusing on return portfolio

⁹Conrad, Hameed, and Niden (1994) sort on changes in the number of transactions. Cooper (1999) sort on changes in turnover. We elect to sort on turnover because the CGW model relates the serial correlation patterns in returns to the level and not changes in turnover (see equation (16) in CGW). Also, the empirical analysis in CGW uses the detrended level of turnover and not changes in turnover. Conrad, Hameed, and Niden (1994) note that their results are unchanged when sorting on changes in trading volume instead of changes in the number of transactions. With two-way sorts on returns and changes in turnover, we find that amongst the loser stocks the low change in turnover securities revert more (a return of 0.64% in week t) than the high change in turnover securities (a return of 0.25%).

one, we note that the returns of each of the sixteen, illiquidity-turnover sorted portfolios is positive, consistent with a reversal in returns. Except for illiquidity portfolio one, the return difference between the turnover portfolio four and portfolio one is positive and significant. This return difference increases monotonically across the illiquidity portfolios from an insignificant 0.08% to 0.68% per week, suggesting that reversals increase with turnover and are much more pronounced when illiquidity is higher. Similarly, the return difference between illiquidity portfolios four and one increase across the turnover portfolios from 0.26% to 0.75%.¹⁰

Let us gauge more specifically the impact of illiquidity on reversal profitability in return portfolio 1. Note that in the highest turnover portfolio, returns increase from 0.42% for the lowest illiquidity portfolio to 1.16% for the highest illiquidity portfolio. This 74 basis-point per week difference can be attributed to illiquidity.

Moving to return portfolio four, we demonstrate that 13 of the 16 returns are negative albeit significantly so in three cases only. Also, the reversal increases with illiquidity only for the highest turnover portfolio and with turnover only for the two high illiquidity portfolios. There is no reversal in return portfolio three. Return portfolio two does have reversals and the reversals increase with turnover (illiquidity) across the illiquidity (turnover) portfolios. Panel B of Figure 1 illustrates the reversals in the extreme return portfolios.

To summarize, in the extreme return portfolios (especially the loser portfolios), reversals increase with turnover and illiquidity. In addition, the high volatility of the extreme return and the high turnover / illiquidity portfolios suggests that it is the non-informational trades that generate the price movements in these portfolios. The findings thus suggest that there are liquidity stresses in the high turnover and high illiquidity portfolios and reversals occur as these liquidity stresses caused by non-informational trades are accommodated by suppliers of liquidity.

3.1.1 Robustness Checks

At this stage, we must ensure that our results are robust to various important considerations.

¹⁰Note that the return differences across the portfolios are not the same as the differences in the average returns of the two portfolios because some of the portfolios have no stocks during a given week. We will provide more details about the number of stocks in each portfolio in section 3.1.1.

Number of stocks in each portfolio: Since we are following a strategy of independently sorting stocks into return-turnover-illiquidity portfolios, it is possible that some of these portfolios may end up with a few or no stocks. Recall that we need a stock to trade every day of the week before it is included in the sample for that week. Not all stocks trade every day. This lack of trading along with independent sorting can result in some portfolios with no stocks during some weeks. We find that the most problematic portfolio in Table 1 is the loser portfolio with the lowest turnover and the lowest illiquidity. For this portfolio, out of the possible 2,111 weeks, 562 weeks have no stocks. The other problematic portfolio is the winner portfolio with the lowest turnover and the lowest illiquidity. For this portfolio, 336 weeks have no stocks. Given that most of the reversal takes place in the high turnover, high illiquidity portfolios, these missing weeks are not problematic for our reversal results. The high turnover, high illiquidity loser portfolio has only four weeks with no stocks and the high turnover, high illiquidity winner portfolio has only two weeks without stocks. Below, we also confirm our results using the relative strength strategies that, by design, do not have any missing weeks.¹¹

Non-synchronous trading: Non-synchronous trading can lead to erroneous stock return autocorrelations for two reasons. The first is attributable to misclassification of stocks due to non-trading during the formation week $t - 1$ and the second is attributable to mismeasurement in the week t .

Consider the misclassification error first and imagine a situation in which the market declines strongly in the first two days of week $t - 1$ and then rises strongly over the rest of the week. If a stock (with a positive beta) traded only during the first two days of the week and had large negative returns it would be classified as a loser stock even though had it traded later in the week it would have recorded positive returns. Even if the market return is zero over week t the stock return will be strongly positive if it trades during week t . This situation would result in a reversal. Thus, non-synchronous trading can lead to misclassification errors and stock returns could erroneously exhibit negative serial correlations.

Let us now turn to the mismeasurement in the week t . Consider a situation in which the market return is negative in week $t - 1$; the return is positive during the first two

¹¹We have also implemented further analysis by sorting sequentially (instead of independently). This ensures that, by design, all portfolios are fully and equally populated. The results (available upon request) show that the potential reversal profits are very similar to those reported. After controlling for turnover, the returns increase with illiquidity. Consider, for example, turnover portfolio one (four). The returns to the loser minus winner strategy increase from 0.36% (0.48%) to 0.53% (1.18%), as illiquidity increases from the lowest to highest.

days of week t but becomes negative over the rest of week t so that the overall return in week t is negative. If a stock trades on all days of week $t - 1$ and only on the first two days of week t then it will be correctly classified as a loser stock in week $t - 1$. This stock will exhibit negative serial correlation even if it would not have exhibited a reversal in returns had it traded on the other days of week t . Thus, non-synchronous trading could erroneously result in reversals.

Our experimental design ensures that our results are not being contaminated by non-synchronous trading. Recall that our analysis uses only stocks that traded on each and every day of the week to be included in that week. More specifically, a stock that did not trade on any day of a given week was dropped from the sample for that week and the following week to ensure that there would be no misclassification or mismeasurement errors. Moreover, it is typically the low priced stocks that have periods with no trading. Recall that we have eliminated penny stocks from the sample. In addition, we reran our entire analysis using only stocks with prices larger than \$5. Qualitatively, the results remain unchanged.

In sum, non-synchronous trading is not the source of return reversals in individual stocks at the weekly frequency. This conclusion is consistent with Foerster and Keim (2000) who report that the likelihood of a NYSE-AMEX stock going without trading for two consecutive days is 2.24%. Of course, it is possible that non-trading within the day could lead to misclassification and mismeasurement errors, but we do not expect these high frequency intra-day non-trading issues to be severe enough to impact serial correlations at the weekly or monthly frequency.

Seasonality: We also checked for the presence of seasonalities in the serial correlation patterns of returns. Upon replicating Table 1, with all weeks in January eliminated, we find that the results are practically unchanged. For instance, in the highest illiquidity portfolio the difference in returns between the high turnover and the low turnover stocks is -0.33% compared to -0.31% in Table 1. In the highest illiquidity portfolio the difference in returns between the high turnover and the low turnover stocks is 0.60% compared to 0.68% in Table 1.

Alternative measures of illiquidity: Thus far we have focused on the Amihud illiquidity measure. We have checked that our results are robust to different measures of illiquidity. Following Chordia, Roll, and Subrahmanyam (2001) we use average daily spreads computed from transactions data (ISSM and TAQ) as a measure of illiquidity

over the sample period 1988 through 2002.¹² As in Chordia, Roll, and Subrahmanyam (2001), only NYSE stocks are included in the sample and only the bid and ask quotes on the NYSE are used to calculate the spreads. In addition, the sample selection, quote and trade matching, and filtering rules are the same as in Chordia, Roll, and Subrahmanyam (2001). The weekly illiquidity measures are the averages of the daily measures during the week. The different measures of illiquidity that we use are: (i) proportional quoted spreads, (ii) proportional effective spreads, (iii) proportional effective spreads divided by the depth (average of the ask and the bid quoted trade sizes), and (iv) proportional effective spreads divided by the depth.

Moreover, Ball, Kothari, and Wasley (1995) have argued that it is very important to control for the bid-ask bounce when computing the profitability of short-run trading strategies. While we continue to use the skip day methodology, we also follow Conrad, Hameed, and Niden (1994) in calculating the daily returns from closing prices at the mid-point of the bid-ask spread.¹³

Thus, we check for robustness of our results along three dimensions: (i) we use different measures of illiquidity, (ii) we use mid-point returns along with the skip day methodology to eliminate the bid-ask bounce, and (iii) we conduct this analysis for the subperiod 1988-2002 for which the transactions data are available.

In untabulated results (available upon request), we find that the empirical findings are robust along the above three dimensions. The results are essentially the same as those based on the Amihud measure when using the different illiquidity measures. Some of the returns are lower (though still statistically and economically significant) but so are the standard deviations. We observe strong reversals in the high turnover, high illiquidity, loser portfolios regardless of how illiquidity is measured. In addition, over the 1988-2002 period for NYSE stocks there are no reversals in the winner stocks even when illiquidity is as per the Amihud measure.

3.1.2 Are the high turnover, low liquidity stocks really different?

Table 1 and Figure 1 show that stocks facing the most price pressure in week $t - 1$ reverse the most in week t . However, from only this evidence it is unclear whether we

¹²This data sample is not the same as in the rest of the paper, but does provide alternative measures of illiquidity.

¹³See Chordia and Subrahmanyam (2004) for details about calculating returns from closing mid-point prices.

learn anything more than the fact that individual stocks have negative serial correlations at the weekly frequency. Given the negative serial correlation, it may not be surprising that stocks that have higher (lower) returns in week $t - 1$ will, on average, have lower (higher) returns in week t . Are the high turnover, highly illiquid stocks really different? In particular, do these stocks reverse more than the low turnover, more liquid stocks?

To address that question, we estimate cross-sectional regressions similar to those in Jegadeesh (1990). Specifically, we estimate the following regression for various turnover and/or liquidity groups at the weekly frequency:

$$R_{it} = \alpha_t + \beta_t R_{it-1} + \epsilon_{it} . \quad (2)$$

If the various turnover/liquidity groups are not different then the slope coefficient β in the above equation will be similar across the groups.

Table 2 presents the Fama-Macbeth coefficients from the cross-sectional regressions. The time-series average of β based on all stocks in the sample is -0.050 confirming the fact that in the cross-section stocks with higher (lower) returns in week $t - 1$ have lower (higher) returns in week t . Panel B (C) presents the coefficients for stocks sorted by turnover (illiquidity). The average β for the lowest turnover (illiquidity) stocks is -0.038 (-0.042), while for the highest turnover (illiquidity) stocks it is -0.051 (-0.062). Thus, high turnover, less liquid stocks are more negatively autocorrelated than the low turnover, more liquid stocks. The same pattern is observed in Panel D when simultaneously sorting on turnover and illiquidity. Consider the highest turnover stocks. The average beta from equation 2 increases monotonically from -0.035 to -0.072 with illiquidity. This difference is significant at the 1% level. A similar increase with turnover is observed for the most illiquid stocks. For instance, for the highest illiquidity portfolio the betas increase from -0.043 to -0.072 with turnover. Once again this difference is statistically significant at the 1% level.

We conclude that high turnover, low liquidity stocks are indeed different from low turnover, high liquidity stocks in their reversal patterns – they are more negatively autocorrelated. Thus, the reversals in Table 1 do not simply represent one uniform negative serial correlation. The reversals obtain *both* because the high turnover, low liquidity stocks face the most price pressure in week $t - 1$ *and* because those stocks have the most negative serial correlations.

If the negative serial correlation patterns in individual stocks are more pronounced for

stocks with higher turnover and higher illiquidity then we should observe higher reversals with strategies that overweight the high turnover and the high illiquidity stocks. Next, we study the relative strength strategies.

3.2 Relative Strength Strategies

Thus far, we have calculated equally weighted average returns for each of our portfolios. We now follow the spirit of the relative strength strategy of Lehmann (1990), Lo and MacKinlay (1990), and Conrad, Hameed, and Niden (1994). That strategy puts heavier weights on stocks with the extreme returns, turnover, and illiquidity. In the end, the relative strength strategy provides an important metric upon which to assess the incremental impact of illiquidity on reversal profitability.

We focus on the three-way sorts based on returns, turnover, and illiquidity.¹⁴ The weight on stock i in week t for each of the eight portfolios is as follows:

$$\begin{aligned} w_{pit} &= \frac{R_{it-1}T_{it-1}L_{it-1}}{\sum_{i=1}^{N_p} R_{it-1}T_{it-1}L_{it-1}}, \\ p &= WHH, WHL, WLH, WLL, LHH, LHL, LLH, LLL, \end{aligned} \quad (3)$$

where T_{it-1} (L_{it-1}) represents the turnover (illiquidity) of stock i in week $t - 1$ less the corresponding cross-sectional median of turnover (illiquidity) in week $t - 1$, and N_p represents the number of stocks in each of the eight portfolios formed by the three-way sort between winners (W) and losers (L), and high (H) and low (L) turnover and illiquidity. The results are presented in Table 3 (weekly frequency) and Table 4 (monthly frequency) for three way sorts and different weighting criteria, i.e., by weighting on returns; returns and turnover; returns and illiquidity; and returns, illiquidity, and turnover. The two-way weighting criterion provides an idea of how important turnover and illiquidity are as compared to just weighting on returns. Moreover, it enables us examine the importance of illiquidity vis-a-vis turnover.

3.2.1 Weekly Frequency

Consider the first set of results for winner stocks with high turnover. For these stocks, high illiquidity results in considerable reversal of returns regardless of the weighting

¹⁴Results on one- and two-way sorts are available upon request.

criterion. For instance, the week t return is -0.85% when the weighting criterion is based on returns, turnover, and illiquidity as in equation (3). The return for the low illiquidity portfolio is insignificantly different from zero for the same weighting criterion. A zero investment winner and high turnover portfolio that is long stocks with high illiquidity and short stocks with low illiquidity shows significant reversals. For example, weighting by all three characteristics results in a zero investment portfolio return of -0.78% per week. Next, depending on the weighting criterion, a zero investment portfolio of high illiquidity winners that is long the low turnover stocks and short the high turnover stocks has a week t return that ranges from 0.43% to 1.00% with standard deviations ranging from 1.9% to 6.7% .

Let us move to loser stocks. The documented reversals are all statistically significant and are typically larger than those for the winner stocks. High turnover loser stocks have higher reversals than the low turnover loser stocks. With the weighting criterion as in equation (3), the zero investment portfolio composed of high (low) turnover loser stocks, that is long on the high illiquidity and short on the low illiquidity stocks has a week t return of 1.50% (0.69%). High illiquidity loser stocks also have higher reversals than the low illiquidity loser stocks. With the weighting criterion as in equation (3), the zero investment portfolio composed of high (low) illiquidity loser stocks, that is long on the high turnover and short on the low turnover stocks has a week t return of 0.89% (0.09%).

Indeed, the different weighting criteria agree that both illiquidity and turnover contribute to the reversal in returns. Moreover, as illustrated below, the impact of illiquidity seems to be larger. Overall, our findings provide strong support to the idea that illiquidity provides additional insights (over and above that obtained from turnover alone) for understanding the nature of short-term reversal profitability.

To start, when weighting by returns and turnover, the high illiquidity, high turnover, loser portfolio has weekly return of 1.19% . However, when weighting by returns and illiquidity the same portfolio has a weekly return of 1.69% . Note also that the zero investment high turnover loser portfolio that is long on high illiquidity stocks and short on low illiquidity stocks has a return of 1.28% (0.83%) when the weighting criterion is returns plus illiquidity (returns plus turnover). In fact, except for the low turnover winner portfolio, if we follow the two columns for the zero investment portfolios where the weighting criteria are returns plus turnover and returns plus illiquidity this pattern of more reversals when weighting by illiquidity instead of turnover is repeated throughout this table.

Moreover, the highest profits from a long-short portfolio is obtained in the high turnover, high illiquidity stocks when going long the loser stocks and short the winner stocks. When the weighting scheme is as in equation (3) the weekly return from this strategy is 2.69%. When considering the different weighting scheme in the different columns the results suggest that weighting by illiquidity has a larger impact than weighting by turnover. For instance, when weighting by return and turnover, the loser minus winner long-short portfolio has a weekly return of 1.78% but when weighting by return and illiquidity the return is 2.21%, as compared to 1.43% when weighting by returns only. Also, regardless of the weighting scheme, the loser minus winner portfolio has higher returns whenever the sample consists of high illiquidity stocks. Thus, once again, the results suggest that illiquidity is an important driver of the short-term reversals of individual stock returns.

The incremental impact of illiquidity can be ascertained in a number of ways. One way is to take the difference between 1.83% and 1.19% for the loser high turnover, high illiquidity portfolio (L,H,H), which yields 64 basis points per week. Another way is to take the difference between 1.50% and 0.83% (L,H,H-L), yielding 67 basis points per week. One can also look at the strategy (L-W,H,H). Here the illiquidity impact is 91 [2.69%-1.78%] basis points per week. That is, regardless of how we slice the data, illiquidity does have a significant impact on reversal strategy profits. Each of these three strategies demonstrates the statistically and economically significant impact of illiquidity, over and above that of turnover, ranging between 64 to 91 basis points per week.

Before we conclude this section, let us note that reversal patterns are weaker when the week t stock returns are equally weighted. In fact, there are almost no reversals in the winner portfolios. There are reversals in the loser portfolios but the reversals are weaker than those based on the other weighting schemes. Also the return volatilities with an equal weighting scheme are substantially smaller. Observe that the results from the relative strength strategies are stronger than those presented in Table 1 when the portfolios are formed by equally weighting the stocks within the various portfolios. These findings support our argument that the return reversals are stronger for stocks with extreme returns, trading volumes, and illiquidity values.

3.2.2 Monthly Frequency

All our analysis thus far is confined to weekly returns. Since Jegadeesh (1990) has documented negative serial correlations in the cross-section of monthly returns, we now focus on the monthly frequency to check whether the impact of turnover and illiquidity continues to hold. The results are presented in Table 4 which is the monthly equivalent of Table 3.

Starting with the winner portfolios, there is reversal in the high turnover winners with high illiquidity but only when the weighting criterion includes illiquidity. The high turnover, low illiquidity winners exhibit continuations regardless of the weighting criterion. The same result obtains in the case of the low turnover winners. With the weighting criterion based on returns and illiquidity, a zero investment high (low) turnover, winner portfolio that is long on low illiquidity stocks and short on high illiquidity stocks has an average monthly return of 1.79% (1.64%). The reversals are far lower with the weighting criterion based on returns and turnover. Moreover, there is no reversal in the zero investment high (low) liquidity winner portfolio that is long on high turnover stocks and short on low turnover stocks. Thus, at the monthly frequency, turnover does not affect the serial correlation in the winner portfolios.

Let us turn to the loser portfolios. With the weighting criterion as in equation (3), the zero investment high turnover, loser portfolio that is long the high illiquidity stocks and short the low illiquidity stocks has a return of 2.49%. The high turnover stocks exhibit more continuations than the low turnover stocks for the low illiquidity losers.

Next, the loser minus winner portfolio exhibits higher profitability for the more illiquid stocks. The profitability is the lowest when the weighting criterion is based on returns and turnover and the highest when based on returns and illiquidity. For instance, the loser minus winner profitability is 3.70% (0.68%) when the weighting criterion is based on returns and illiquidity (returns and turnover). In fact, the profitability based on weighting by returns and turnover is lower than that based on returns alone suggesting that conditioning on turnover hurts the profitability of the loser minus winner portfolio.

Finally, let us assess the marginal impact of illiquidity, controlling for turnover, on reversal profitability. Based on the (L,H,H), (L,H,H-L), and (L-W,H,H) strategies the illiquidity impact is 1.82% [2.86%-1.04%], 1.87% [2.49%-0.62%], and , 2.84% [3.52%-0.68%] per month, respectively, Once again, regardless of how we slice the data, illiquidity does have a significant impact on reversal strategy profits also at the monthly

frequency.

To summarize, at the monthly frequency the negative serial correlation results are weaker for sorts based on turnover suggesting that turnover may be a poor proxy for the liquidity demands of the non-informational traders at the monthly frequency. Recall that at the weekly frequency these non-informational demands did lead to reversals, as suggested by CGW. Contrary to the results for turnover, the impact of liquidity on the serial correlation patterns of individual stock returns, continues to be strong at the monthly frequency. This is consistent with the conjecture that stresses in the market for liquidity lead to large price movements which are subsequently reversed as liquidity suppliers step in to profit from prices that diverge from fundamental values.

4 Transaction Costs

Table 1 shows that the potential profits from a short-run contrarian strategy can be large. For instance, in the case of the extreme loser portfolio with high turnover and high illiquidity, the equally weighted contrarian strategy return is 1.16% per week. However, this portfolio has the highest volatility (4.37% per week) and, since this is the highest illiquidity portfolio, we suspect that transaction costs may render any trading strategy unprofitable. We now examine transaction costs. Specifically, we first provide descriptive statistics about the various trading strategies considered here, we then use the transaction cost estimates of Keim and Madhavan (1997) within a dynamic framework to estimate net returns based on a strategy that buys the losers and sells the winners, and we finally consider the market impact cost analysis of Korajczyk and Sadka (2004).

4.1 Portfolio Characteristics

Table 5 presents descriptive statistics about the various three-way sorted portfolios in Table 1.¹⁵ Controlling for turnover, the weekly dollar trading volume decreases monotonically with illiquidity. The dollar trading volume for the lowest illiquidity portfolio is orders of magnitude larger than that of the highest illiquidity portfolio. For instance,

¹⁵These descriptive statistics are for the post-formation week. At least part of the round-trip trading costs are incurred in the formation week and it is perhaps more relevant to check the descriptive statistics in the formation period. However, by design, the differences in the high and low turnover or high and low liquidity are even more stark in the formation periods and, as such, we are being conservative in our estimates of the transaction costs by basing them on the statistics from the post-formation week.

for the extreme loser portfolio with the highest turnover, the dollar trading volume for the lowest (highest) illiquidity portfolio is \$17.79 million (\$0.13 million) per week. Consequently, the illiquidity measure in the lowest quartile is 0.03 and that for the highest quartile is 3.01, which is a hundred times higher. The differences in turnover are not as stark, especially for the higher turnover portfolios. In sum, the high illiquidity portfolios have trading volumes that are, in general, orders of magnitude smaller than those of the low illiquidity portfolios. The high illiquidity portfolios have stocks in the lowest size quintiles, as opposed to the low illiquidity portfolios that have high trading volumes and stocks in the fourth or fifth highest size quintiles as per NYSE breakpoints.¹⁶

Panel B of Table 5 presents the bid-ask spread and the depths of the various three-way sorted portfolios of NYSE stocks over the period 1988 through 2002. Focusing once again on the extreme loser portfolios with the highest turnover, we note that the proportional effective spread increases from 0.33% for the lowest illiquidity portfolio to 2.35% for the highest illiquidity portfolio. The quoted depth is always lower for the high illiquidity portfolios and the difference is the largest for the highest turnover portfolios. Except for the lowest illiquidity portfolio, the effective spread is larger than the potential trading profits from a contrarian strategy. For instance, the potential profit of 1.16% for the loser portfolio with the highest turnover and illiquidity seems to be swamped by the proportional effective spread of 2.35%.

Table 6 documents the portfolio characteristics for the relative strength strategies with the weighting scheme as in equation (3). The difference between the illiquidity measure between the high and the low liquidity portfolios is far greater than that of Table 5. This is consistent with the evidence that the relative strength strategies place greater weights on stocks that are extreme in terms of returns, liquidity, and turnover. Also, note in Panel B that the proportional spread is larger with the relative strength strategies than with the equally weighted strategies. Recall from Panel C of Table 3 that the profits from buying losers and selling winners was 2.69%. The loser minus winner relative strength portfolio with high turnover and high liquidity has a proportional effective spread of 3.03% suggesting that any potential profits are swamped by the round-trip costs of 6.06%. However, institutional investors carefully watch transactions costs. They choose trade size and spread out their trades over days so as to minimize trading costs. Therefore, the proportional spread of 3.03% may be an overstatement.

To obtain more precise estimates of institutional trading costs, we turn to Keim and

¹⁶NYSE breakpoints were obtained from the data library at Ken French's website. See <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>.

Madhavan (1997) (henceforth KM) estimates of transaction costs.

4.2 KM Estimates of Transaction Costs

First, we present some back-of-the-envelope calculations. KM provide estimates of market impact as well as commission costs for buy and sell trades for NYSE-AMEX stocks (as well as NASDAQ stocks) traded by 21 large institutions during 1991 to 1993. Their transaction cost data is categorized by trade size as well as by firm size. We will focus on the smallest trade sizes to minimize trading costs. We will also use the various size classifications from Table 5. Once again, we focus on the extreme loser portfolio because the potential profits are the highest for these portfolios. Consider the lowest turnover, lowest illiquidity portfolio that has a contrarian trading profit of 0.31% per week (see Table 1). The stocks in this portfolio belong to the largest size quintile. According to KM, the cost of a small buy trade in the largest size quintile is 0.31%, and the cost of a small sell trade in the same size quintile is 0.21% for a total round-trip cost of 0.52%. This cost is larger than the potential profit of 0.31%. Since most of the potential trading profits in the low illiquidity portfolios are lower than 0.52%, we now turn to the high illiquidity, high turnover portfolio. The average size quintile corresponding to this portfolio is 1.48. Interpolating between the two lowest size quintiles we find that cost of a small buy trade is 0.38%, and the cost of a small sell trade is 0.88% for a total round-trip cost of 1.26%. This cost, once again, is larger than the potential profit of 1.16%.

The caveat with the above analysis is that the data in KM pertains to institutional trades during 1991-1993. Trading costs are likely to be higher during the early part of the sample. Also, trading costs have not declined over the latter part of the sample.¹⁷ We will now conduct a dynamic analysis of the trading costs by using the methodology in Cooper, Gutierrez, and Marcum (2005). This methodology applies the trading costs estimates from Stoll (1995) to the sample specific estimates of KM.¹⁸ In order to be conservative, we will (i) estimate costs for value traders since they have the lowest trading costs, (ii) set the trade size to zero, and (iii) place a cap of 2.5% on the trading costs for each stock. The results are presented in Tables 7 and 8 for the equally weighted portfolios and the relative strength portfolios, respectively.

Consider the equally weighted portfolio strategies in Table 7. The highest potential

¹⁷Personal communication with Don Keim.

¹⁸For details, see the appendix in Cooper, Gutierrez, and Marcum (2005).

profits occur in the high turnover, high illiquidity portfolios. The loser minus winner long-short strategy results in raw weekly returns of 1.33%. However, due to low liquidity, the net return after accounting for transactions costs is -5.26%. Transactions costs overwhelm the potential profits. This is true of all the turnover-illiquidity portfolios except for the low turnover, low illiquidity portfolio. Even for this portfolio the weekly returns after transactions costs is only eleven basis points. Similarly, from Table 8 we see that the trading costs for all turnover-illiquidity portfolios are far larger than the potential profits for the relative strength strategies. The potential weekly return of 2.69% for the high turnover, high illiquidity portfolio turns into a net return of -4.48% after accounting for transactions costs.

While it seems that the above transaction costs are large, recall that we have been quite conservative in estimating these costs. Even so, the costs may actually be understated even more due to the following reasons:

1. KM use data for trades that were actually consummated. If some trades were abandoned due to high transactions costs then the KM estimates of trading costs would be biased downwards.
2. The transaction cost estimates in KM do not include short selling costs.

To provide yet another dimension of transaction costs, we also implement the analysis of Korajczyk and Sadka (2004).

4.3 Korajczyk and Sadka Estimates of Market Impact Costs

Korajczyk and Sadka (2004) use intraday transactions data to estimate market impact costs. These costs are estimated each month for each stock over the period January 1993 through May 1997. Using cross-sectional relationships between these market impact costs and firm characteristics, the costs are then estimated for the entire sample period.¹⁹ Since the costs are estimated for each stock each month, we keep them constant for the month in our weekly analysis. The estimated market impact costs are then used to compute the cost of the taking positions in the extreme winner or loser stocks each week.²⁰ The results are presented for \$10,000 of initial investment (equivalent to \$60,000

¹⁹We thank Ronnie Sadka for providing us this data.

²⁰For details, see Appendix A in Korajczyk and Sadka (2004).

in 2002 dollars) in Tables 7 and 8 for the equally weighted portfolios and the relative strength portfolios, respectively.

From Table 7 we see that the highest potential weekly return of 1.33% from the high turnover, high illiquidity, loser minus winner strategy turns into a net return of -1.09% when the market impact costs are accounted for. In fact, all the net returns are negative. The same result obtains for the relative strength strategies in Table 8. The high turnover, high illiquidity, loser minus winner portfolios have the highest potential returns of 2.69% and net returns of -0.20%. Thus, an investment portfolio as small as \$60,000 in 2002 dollars has enough market impact costs to swamp all potential profits from a strategy that exploits the short-term reversal phenomenon in individual stock returns. While, it is important to consider long-short trading strategies to eliminate market risk, it should be pointed out that market impact costs (with a \$10,000 initial investment) are not sufficient to eliminate all the potential profits from going long in the high turnover, high illiquidity stocks. A net return of 24 basis points obtains after accounting for market impact costs. Even so, these costs understate the true transactions costs because they do not account for commissions. In Table 2 of their paper, KM show that commissions for institutional investors can be more than 60% of the market impact costs. Note also that the analysis of Korajczyk and Sadka (2004) does not account for short sale costs.

4.4 Summary of Transactions Costs

This paper analyzes trading costs from a number of different perspectives, including proportional effective bid-ask spreads, dynamic institutional trading costs, and the market impact costs. Consider the relative strength strategy, loser minus winner portfolio with the highest turnover and the highest illiquidity. The round-trip proportional bid-ask spread cost for this portfolio calculated over the sample period 1988 through 2002 is 6.06%. The dynamic cost analysis of institutional trading costs using the methodology of Cooper, Gutierrez, and Marcum (2005) suggests that transactions costs including commissions are 7.17%. The market impact costs (not including commission costs) using the Korajczyk and Sadka (2004) methodology amount to 2.89% for an investment portfolio as small as \$10,000. All these costs are larger than the potential contrarian trading strategy profits.

The overall conclusion from an analysis of bid-ask spreads, market impact costs, and institutional transactions costs is that the potential contrarian trading profits are not attainable after accounting for trading costs. The main reason is that the largest

potential profits are obtained in the highly illiquid stocks that have high trading costs.

5 Conclusions

Stock returns can be predictable at low frequencies due to time variation in expected returns that is consistent with rational asset pricing. However, predictability at high frequencies, such as stock return reversals, poses a serious challenge to the efficient market hypothesis, because stock prices are likely to follow a martingale process over short horizons. The notion is that short-run changes in fundamental values should be negligible in efficient markets with random information arrival. High frequency price reversals have been explained within a rational equilibrium paradigm by CGW (1993). The CGW model is analogous to the inventory cost models of market making. The idea is that demand shocks from non-informational traders is absorbed by market makers who require compensation for holding non-optimal inventories. This compensation is provided in the form of short-run reversals. In the model, trading volume plays an important role in identifying price changes due to either public information or exogenous selling pressure by non-informational trades. Non-informed trading is accompanied with high trading volume, whereas informed trading, which does not lead to reversals, is accompanied with little trading volume. Thus, price changes accompanied with high (low) trading volume should (should not) revert.

The CGW paradigm assumes downward sloping demand curves for stocks. If demand curves were perfectly elastic, there would be no impact of trades on prices and subsequent price reversals would not occur. With downward sloping demand curves, however, price reversals should follow liquidity or non-informational trading. Stocks that are more illiquid should experience stronger reversals. Consequently, this paper incorporates the unexplored dimension of illiquidity into the empirical investigation of the serial correlation patterns of individual stock returns. Indeed, we show that liquidity plays an important role in understanding the autocorrelation patterns in stock returns.

Here is a summary of our findings: (i) There is reversal in weekly and monthly stock returns but it is mainly confined to the loser stocks; (ii) At the weekly frequency, high turnover stocks exhibit higher negative serial correlations than low turnover stocks; (iii) At the monthly frequency the impact of turnover on autocorrelations is the opposite of that at the weekly frequency – low turnover stocks exhibit more reversals than high turnover stocks; (iv) Controlling for turnover, there is more reversal in stocks with

low liquidity than in the highly liquid stocks and this pattern is the same across the weekly and the monthly frequency; (v) The high turnover, low liquidity stocks have more negative serial correlations in cross-sectional regressions, and the potential contrarian trading strategy profits are the highest for these stocks; (vi) The largest potential profits before accounting for transactions costs obtain in the high turnover, high illiquidity, loser minus winner stocks.

These findings accord with the hypothesis that extreme price changes occur in stocks with low liquidity and in the high turnover stocks, where the demand for liquidity from uninformed traders is high. The largest price reversals occur in precisely these high volume, low liquidity stocks as the initial price changes are reversed. Thus, the high frequency negative autocorrelations are more likely to result from stresses in the market for liquidity. A high frequency trading strategy that attempts to profit from the negative serial correlations has high transactions costs and substantial price impact. Consequently, while the presence of negative autocorrelations in individual security returns is undeniable, it is not possible to profit from this short-run predictability. This lack of profitability and the fact that the overall findings are consistent with the rational paradigm of CGW (1993) suggest that the violation of the efficient market hypothesis due to short term reversals is not so egregious after all.

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Table 1: Three-Way Sorted Portfolios: Post-Formation Returns

This table presents means and standard deviations in the post-formation week. Equal-weighted portfolios are formed every week. The sorts are based on returns, turnover, and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Return breakpoints are determined by the median return for both positive and negative lagged returns. Breakpoints for turnover and illiquidity are based on quartile breakpoints. Return portfolio 1 is the extreme loser portfolio and return portfolio 4 is extreme winner portfolio. Turnover (illiquidity) portfolio 1 has the lowest turnover (illiquidity), and portfolio 4 has the highest turnover (illiquidity). Portfolio returns are in percent per week and are based on skip-day methodology in which the return on the last day of the week is not used in computations. Stars indicate significance of means at 99%, 95%, and 90% level. The sample period is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for all days of the week.

Turnover	Return Portfolio=1					Return Portfolio=2				
	Illiquidity					Illiquidity				
	1	2	3	4	4-1	1	2	3	4	4-1
1	0.31*** 2.71	0.28*** 2.49	0.27*** 2.33	0.48*** 2.59	0.26*** 2.84	0.08** 1.91	0.08** 1.67	0.09** 1.75	0.15*** 2.14	0.07* 1.89
2	0.38*** 2.56	0.41*** 2.53	0.42*** 2.63	0.66*** 2.95	0.27*** 2.66	0.16*** 1.90	0.20*** 2.12	0.21*** 2.35	0.23*** 2.86	0.07* 2.29
3	0.42*** 2.59	0.48*** 2.80	0.50*** 2.99	0.89*** 3.46	0.46*** 2.87	0.22*** 2.20	0.25*** 2.45	0.25*** 2.84	0.38*** 3.99	0.15** 3.46
4	0.42*** 3.09	0.53*** 3.22	0.73*** 3.57	1.16*** 4.37	0.75*** 3.74	0.29*** 2.63	0.27*** 3.02	0.37*** 3.50	0.50*** 5.90	0.24** 5.50
4-1	0.08 2.77	0.25*** 2.44	0.46*** 2.52	0.68*** 3.46	0.82*** 4.25	0.20*** 1.96	0.20*** 2.11	0.29*** 2.59	0.41*** 5.46	0.42*** 5.67
Turnover	Return Portfolio=3					Return Portfolio=4				
	Illiquidity					Illiquidity				
	1	2	3	4	4-1	1	2	3	4	4-1
1	0.07** 1.91	0.09*** 1.57	0.16*** 1.62	0.24*** 2.10	0.17*** 1.81	-0.01 2.47	-0.10** 1.97	-0.03 1.87	0.14*** 2.28	0.16*** 2.55
2	0.15*** 1.85	0.16*** 1.91	0.20*** 2.17	0.35*** 2.76	0.20*** 2.16	-0.05 2.01	-0.06 2.18	-0.02 2.29	0.10** 2.68	0.15*** 2.26
3	0.19*** 2.12	0.26*** 2.36	0.33*** 2.71	0.57*** 3.43	0.39*** 2.82	-0.04 2.15	-0.05 2.37	-0.02 2.54	0.02 3.08	0.06 2.48
4	0.21*** 2.50	0.28*** 2.89	0.37*** 3.26	0.70*** 4.79	0.46*** 4.31	-0.02 2.61	-0.04 2.79	-0.13** 2.92	-0.17** 3.39	-0.14** 2.87
4-1	0.14*** 2.01	0.18*** 2.02	0.21*** 2.46	0.43*** 4.22	0.63*** 4.53	-0.00 2.38	0.04 2.14	-0.09** 2.05	-0.31*** 2.73	-0.21*** 3.33

Table 2: Cross-Sectional Regressions

This table presents the average β in the following weekly cross-sectional Fama-Macbeth type regressions:

$$R_{it} = \alpha_t + \beta_t R_{it-1} + \epsilon_{it}.$$

t -statistics are given in parenthesis below the average coefficient. Separate regressions are run for different categories of stocks. Panel A is for all stocks; Panel B groups stocks by turnover, Panel C groups stocks by illiquidity, and Panel D groups stocks by turnover and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Breakpoints for turnover and illiquidity are based on quartile breakpoints. Return portfolio 1 is the extreme loser portfolio and return portfolio 4 is extreme winner portfolio. Turnover (illiquidity) portfolio 1 has the lowest turnover (illiquidity), and portfolio 4 has the highest turnover (illiquidity). Portfolio returns are in percent per week and are based on skip-day methodology in which the return on the last day of the week is not used in computations. The sample period is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for all days of the week.

Panel A: All				
-0.050 (-30.18)				
Panel B: Turnover			Panel C: Illiquidity	
1	-0.038 (-14.77)		1	-0.042 (-14.57)
2	-0.047 (-19.04)		2	-0.044 (-18.33)
3	-0.060 (-26.20)		3	-0.051 (-23.86)
4	-0.051 (-27.59)		4	-0.062 (-33.12)
Panel D: Turnover and Illiquidity				
Turnover	Illiquidity			
	1	2	3	4
1	-0.045 (-3.89)	-0.047 (-9.76)	-0.038 (-10.90)	-0.043 (-11.81)
2	-0.060 (-13.83)	-0.058 (-16.55)	-0.054 (-16.49)	-0.050 (-14.72)
3	-0.066 (-17.58)	-0.059 (-17.46)	-0.059 (-18.56)	-0.074 (-21.83)
4	-0.035 (-11.03)	-0.042 (-15.34)	-0.056 (-20.22)	-0.072 (-23.49)

Table 3: Relative Strength Portfolios: Post-Formation Returns

This table presents portfolio returns in the post-formation week. Portfolios are formed every week. The sorts are based on returns, turnover, and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Return breakpoints are determined by the median return for both positive and negative lagged returns. Breakpoints for turnover and illiquidity are based on quartile breakpoints. Weights on stock i in week t are based on

$$w_{pit} = \frac{R_{it-1}T_{it-1}L_{it-1}}{\sum_{i=1}^{N_p} R_{it-1}T_{it-1}L_{it-1}},$$

$$p = WHH, WHL, WLH, WLL, LHH, LHL, LLH, LLL,$$

where T_{it-1} (L_{it-1}) represents the turnover (illiquidity) of stock i in week $t - 1$ and N_p is the number of stocks in each of the eight portfolios formed by the three-way sort between winners (W) and losers (L) and high (H) and low (L) turnover and illiquidity. The results are presented for different weighting criteria. Returns are in percent per week and are based on skip-day methodology in which the return on the last day of the week is not used in computations. Stars indicate significance of means at 99%, 95%, and 90% level. The sample period is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for all days of the week.

Sorting Criterion			Weighting Criterion				
Return	Turnover	Illiquidity	Return	Return Turnover	Return Illiquidity	Return Turnover Illiquidity	equal weight
W	H	H	-0.43** 2.89	-0.58** 3.75	-0.62** 5.00	-0.85** 6.69	0.09* 2.59
W	H	L	-0.09* 2.47	-0.09 2.88	-0.07 2.44	-0.07 2.86	0.08 2.28
W	L	H	0.00 2.17	0.03 2.09	0.15** 3.32	0.15** 3.28	0.16** 1.93
W	L	L	-0.05 1.81	-0.04 1.71	-0.04 1.81	-0.04 1.70	0.07* 1.66
W	H	H-L	-0.35** 1.89	-0.49** 3.15	-0.55** 4.51	-0.78** 6.40	0.02 1.18
W	L	H-L	0.05 1.43	0.07** 1.40	0.19** 2.94	0.19** 2.90	0.09** 0.97
W	H-L	H	-0.43** 1.88	-0.61** 3.09	-0.77** 4.80	-1.00** 6.66	-0.06** 1.13
W	H-L	L	-0.04 1.38	-0.06 2.03	-0.03 1.40	-0.04 2.03	0.01 0.98
L	H	H	0.99** 3.32	1.19** 4.72	1.69** 5.26	1.83** 6.69	0.65** 2.94
L	H	L	0.43** 2.78	0.37** 3.30	0.40** 2.75	0.33** 3.27	0.36** 2.53
L	L	H	0.57** 2.44	0.52** 2.34	0.99** 3.26	0.94** 3.23	0.33** 2.16
L	L	L	0.29** 2.00	0.25** 1.91	0.28** 1.99	0.24** 1.90	0.19** 1.83
L	H	H-L	0.55** 1.96	0.83** 3.94	1.28** 4.66	1.50** 6.27	0.29** 1.31
L	L	H-L	0.28** 1.46	0.27** 1.45	0.71** 2.68	0.69** 2.72	0.14** 1.11
L	H-L	H	0.42** 1.87	0.68** 3.82	0.70** 4.72	0.89** 6.23	0.32** 1.21
L	H-L	L	0.15** 1.39	0.12** 2.23	0.12** 1.38	0.09* 2.21	0.17** 1.10
L-W	H	H	1.43** 2.29	1.78** 4.61	2.31** 6.08	2.69** 8.52	0.56** 1.20
L-W	H	L	0.52** 1.46	0.46** 2.35	0.47** 1.44	0.40** 2.29	0.28** 0.83
L-W	L	H	0.56** 1.41	0.49** 1.38	0.84** 3.21	0.78** 3.26	0.17** 0.76
L-W	L	L	0.33** 1.01	0.29** 0.95	0.32** 1.08	0.28** 0.97	0.13** 0.64

Table 4: Relative Strength Portfolios: Post-Formation Returns (Monthly Frequency)

This table presents portfolio returns in the post-formation month. Portfolios are formed every month. The sorts are based on returns, turnover, and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Return breakpoints are determined by the median return for both positive and negative lagged returns. Breakpoints for turnover and illiquidity are based on quartile breakpoints. Weights on stock i in month t are based on

$$w_{pit} = \frac{R_{it-1}T_{it-1}L_{it-1}}{\sum_{i=1}^{N_p} R_{it-1}T_{it-1}L_{it-1}},$$

$$p = WHH, WHL, WLH, WLL, LHH, LHL, LLH, LLL,$$

where T_{it-1} (L_{it-1}) represents the turnover (illiquidity) of stock i in month $t - 1$ and N_p is the number of stocks in each of the eight portfolios formed by the three-way sort between winners (W) and losers (L) and high (H) and low (L) turnover and illiquidity. The results are presented for different weighting criteria. Returns are in percent per month and are based on skip-day methodology in which the return on the last day of the month is not used in computations. Stars indicate significance of means at 99%, 95%, and 90% level. The sample period is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for at least ten days of the month.

Sorting Criterion			Weighting Criterion				
Return	Turnover	Illiquidity	Return	Return Turnover	Return Illiquidity	Return Turnover Illiquidity	equal weight
W	H	H	-0.00	0.37	-1.07**	-0.66	0.52
			7.01	8.19	9.11	11.07	6.97
W	H	L	0.69**	0.63**	0.72**	0.68**	0.88**
			5.83	6.73	5.63	6.49	5.53
W	L	H	-0.26	-0.23	-1.19**	-1.06**	0.50**
			5.56	5.43	7.56	7.55	5.29
W	L	L	0.45**	0.45**	0.44**	0.42**	0.70**
			4.27	4.15	4.22	4.12	4.09
W	H	H-L	-0.69**	-0.27	-1.79**	-1.34**	-0.36**
			4.14	5.96	7.70	9.81	3.56
W	L	H-L	-0.71**	-0.67**	-1.64**	-1.49**	-0.19
			3.49	3.47	6.40	6.43	3.01
W	H-L	H	0.26	0.59**	0.12	0.40	0.01
			3.69	5.80	7.30	9.63	2.94
W	H-L	L	0.24*	0.19	0.28**	0.25	0.18
			3.04	4.65	2.94	4.53	2.42
L	H	H	1.36**	1.04**	2.63**	2.86**	1.26**
			8.78	9.71	10.86	12.01	8.06
L	H	L	0.81**	0.42	0.80**	0.37	1.01**
			7.15	8.27	6.95	8.04	6.49
L	L	H	1.20**	1.08**	1.21**	1.09**	1.09**
			6.92	6.61	8.57	8.64	6.13
L	L	L	1.20**	1.12**	1.19**	1.13**	1.12**
			5.06	4.89	4.95	4.82	4.58
L	H	H-L	0.55**	0.62**	1.83**	2.49**	0.25
			4.40	5.85	8.57	9.83	3.68
L	L	H-L	0.00	-0.04	0.02	-0.04	-0.02
			4.31	4.28	6.89	7.14	3.61
L	H-L	H	0.16	-0.04	1.42**	1.77**	0.16
			3.81	5.72	7.43	9.07	3.08
L	H-L	L	-0.39**	-0.70**	-0.39**	-0.77**	-0.11
			3.38	5.15	3.43	5.21	2.87
L-W	H	H	1.36**	0.68**	3.70**	3.52**	0.74**
			4.86	7.09	9.76	11.84	3.14
L-W	H	L	0.12	-0.22	0.08	-0.31	0.13
			3.79	5.41	3.72	5.20	2.36
L-W	L	H	1.46**	1.30**	2.40**	2.16**	0.59**
			3.34	3.40	5.52	5.92	1.93
L-W	L	L	0.75**	0.67**	0.75**	0.71**	0.42**
			2.52	2.58	2.56	2.65	1.67

Table 5: Three-Way Sorted Portfolios: Descriptives

This table presents portfolio characteristics in the post-formation week. Equal-weighted portfolios are formed every week. The sorts are based on returns, turnover, and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Return breakpoints are determined by the median return for both positive and negative lagged returns. Breakpoints for turnover and illiquidity are based on quartile breakpoints. Return portfolio 1 is the extreme loser portfolio and return portfolio 4 is extreme winner portfolio. Turnover (illiquidity) portfolio 1 has the lowest turnover (illiquidity), and portfolio 4 has the highest turnover (illiquidity). The sample period in Panel A is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for all days of the week. Portfolio ‘Sz Rank’ is the average size rank of the stocks in the portfolio, where the size rank (from 1 to 5) is determined from NYSE breakpoints. Portfolio ‘Turnover’ is daily turnover in percent, ‘Volume’ is dollar volume in millions of dollars, and ‘Illiquidity’ is computed using the Amihud measure and is expressed as daily illiquidity (multiplied by 10^6). The sample period in Panel B is 1988 to 2002 and the portfolios include all NYSE stocks that have data for all days of the week. Portfolio ‘Depth’ is the average depth, ‘QSPR’ is the proportional quoted spread, and ‘PESPR’ is the proportional effective spread. Depth is in thousands and proportional spreads are in percent.

Panel A: Nyse/Amex Sample for 1962 to 2002									
Turnover		Return Portfolio=1				Return Portfolio=4			
		Illiquidity				Illiquidity			
		1	2	3	4	1	2	3	4
1	Sz Rank	5.00	4.62	3.65	2.14	4.99	4.56	3.65	2.22
	Turnover	0.082	0.095	0.098	0.096	0.080	0.089	0.084	0.087
	Volume	15.120	1.574	0.410	0.073	14.849	1.436	0.353	0.074
	Illiquidity	0.022	0.088	0.360	5.190	0.034	0.092	0.341	4.481
2	Sz Rank	4.97	4.28	3.13	1.79	4.96	4.26	3.16	1.82
	Turnover	0.153	0.168	0.173	0.163	0.141	0.150	0.149	0.151
	Volume	19.330	1.955	0.476	0.089	19.400	1.825	0.446	0.089
	Illiquidity	0.021	0.099	0.445	3.989	0.026	0.101	0.414	3.652
3	Sz Rank	4.88	3.93	2.81	1.64	4.87	3.93	2.83	1.66
	Turnover	0.238	0.264	0.263	0.236	0.218	0.231	0.230	0.219
	Volume	17.130	2.188	0.545	0.101	16.774	2.070	0.518	0.107
	Illiquidity	0.022	0.114	0.464	3.342	0.026	0.115	0.421	3.072
4	Sz Rank	4.54	3.47	2.45	1.48	4.48	3.39	2.39	1.45
	Turnover	0.597	0.543	0.507	0.404	0.595	0.537	0.479	0.397
	Volume	17.791	2.646	0.644	0.125	16.930	2.557	0.657	0.138
	Illiquidity	0.027	0.133	0.464	3.006	0.029	0.119	0.436	2.572

Panel B: TAQ Sample for 1988 to 2002									
Turnover		Return Portfolio=1				Return Portfolio=4			
		Illiquidity				Illiquidity			
		1	2	3	4	1	2	3	4
1	Depth	7.97	3.84	3.05	3.67	8.37	3.50	2.55	3.13
	PQSPR	0.355	0.623	1.001	2.676	0.364	0.588	0.926	2.328
	PESPR	0.238	0.409	0.660	1.828	0.247	0.385	0.608	1.569
2	Depth	8.06	4.91	4.41	5.29	7.37	4.16	3.62	4.32
	PQSPR	0.405	0.666	1.129	2.801	0.376	0.618	1.025	2.463
	PESPR	0.270	0.439	0.761	1.951	0.251	0.404	0.684	1.701
3	Depth	8.36	6.02	5.87	6.37	7.41	5.11	4.59	5.24
	PQSPR	0.437	0.734	1.284	2.979	0.395	0.674	1.126	2.616
	PESPR	0.291	0.489	0.883	2.100	0.260	0.446	0.759	1.805
4	Depth	9.04	7.66	7.22	8.28	8.36	7.03	6.17	6.77
	PQSPR	0.493	0.847	1.475	3.300	0.456	0.784	1.313	2.855
	PESPR	0.332	0.575	1.029	2.354	0.305	0.529	0.902	2.008

Table 6: Relative Strength Portfolios: Descriptives

This table presents portfolio characteristics in the post-formation week. Portfolios are formed every week. The sorts are based on returns, turnover, and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Return breakpoints are determined by the median return for both positive and negative lagged returns. Breakpoints for turnover and illiquidity are based on quartile breakpoints. Weights on stock i in week t are based on

$$w_{pit} = \frac{R_{it-1}T_{it-1}L_{it-1}}{\sum_{i=1}^{N_p} R_{it-1}T_{it-1}L_{it-1}},$$

$$p = WHH, WHL, WLH, WLL, LHH, LHL, LLH, LLL,$$

where T_{it-1} (L_{it-1}) represents the turnover (illiquidity) of stock i in week $t - 1$ and N_p is the number of stocks in each of the eight portfolios formed by the three-way sort between winners (W) and losers (L) and high (H) and low (L) turnover and illiquidity. The results are presented only for the weighting criteria that utilizes return, turnover, and illiquidity. The sample period in Panel A is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for all days of the week. Portfolio ‘Sz Rank’ is the weighted average size rank of the stocks in the portfolio, where the size rank (from 1 to 5) is determined from NYSE breakpoints. Portfolio ‘Turnover’ is daily turnover in percent, ‘Volume’ is dollar volume in millions of dollars, and ‘Illiquidity’ is computed using the Amihud measure and is expressed as daily illiquidity (multiplied by 10^6). The sample period in Panel B is 1988 to 2002 and the portfolios include all NYSE stocks that have data for all days of the week. Portfolio ‘Depth’ is the weighted average depth, ‘QSPR’ is the proportional quoted spread, and ‘PESPR’ is the proportional effective spread. Depth is in thousands and proportional spreads are in percent.

Panel A: Nyse/Amex Sample for 1962 to 2002						
Sorting Criterion			Descriptives			
Return	Turnover	Illiquidity	Sz Rank	Turnover	Volume	Illiquidity
W	H	H	1.26	0.503	0.145	5.315
W	H	L	3.94	0.881	13.613	0.052
W	L	H	1.63	0.097	0.037	10.551
W	L	L	4.75	0.114	10.736	0.047
L	H	H	1.35	0.444	0.145	6.313
L	H	L	4.09	0.768	13.848	0.057
L	L	H	1.63	0.101	0.036	10.988
L	L	L	4.74	0.121	10.449	0.049
L-W	H	H	1.30	0.473	0.145	5.815
L-W	H	L	4.01	0.825	13.731	0.055
L-W	L	H	1.63	0.099	0.036	10.769
L-W	L	L	4.74	0.117	10.593	0.048
Panel B: TAQ Sample for 1988 to 2002						
Sorting Criterion			Descriptives			
Return	Turnover	Illiquidity	Depth	PQSPR	PESPR	
W	H	H	8.494	3.940	2.795	
W	H	L	9.502	0.597	0.408	
W	L	H	4.625	3.964	2.734	
W	L	L	5.884	0.467	0.310	
L	H	H	9.591	4.501	3.256	
L	H	L	9.472	0.639	0.436	
L	L	H	4.841	4.226	2.968	
L	L	L	6.017	0.501	0.332	
L-W	H	H	9.043	4.220	3.025	
L-W	H	L	9.487	0.618	0.422	
L-W	L	H	4.733	4.095	2.851	
L-W	L	L	5.950	0.484	0.321	

Table 7: Three-Way Sorted Portfolios: Net Returns

This table presents returns net of cost in the post-formation week. Equal-weighted portfolios are formed every week. The sorts are based on returns, turnover, and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Return breakpoints are determined by the median return for both positive and negative lagged returns. Breakpoints for turnover and illiquidity are based on quartile breakpoints. Return portfolio 1 is the extreme loser portfolio and return portfolio 4 is extreme winner portfolio. Turnover (illiquidity) portfolio 1 has the lowest turnover (illiquidity), and portfolio 4 has the highest turnover (illiquidity). Portfolio returns are in percent per week and are based on skip-day methodology in which the return on the last day of the week is not used in computations. The returns are presented for a long position in return portfolio 1 (loser), a long position in return portfolio 4 (winner), and a combined long position in return portfolio 1 and a short position in return portfolio 4. Net returns are calculated by subtracting Keim and Madhavan (1997) and Korajczyk and Sadka (2004) (with a starting dollar amount of 10,000 dollars) cost estimates. These are indicated by the mnemonics ‘KM’ and ‘KS’, respectively. The sample period is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for all days of the week.

Turnover Portfolio	4	4	1	1
Illiquidity Portfolio	4	1	4	1

Return Portfolio=1 (Loser)				
Raw Return	1.16	0.42	0.48	0.31
Net Return (KM)	-2.26	-0.00	-2.22	0.18
Net Return (KS)	-0.18	-0.18	-0.54	0.04

Return Portfolio=4 (Winner)				
Raw Return	-0.17	-0.02	0.14	-0.01
Net Return (KM)	-3.34	-0.40	-2.31	-0.14
Net Return (KS)	-1.24	-0.54	-0.79	-0.32

Return Portfolio=1-4 (Loser-Winner)				
Raw Return	1.33	0.44	0.34	0.40
Net Return (KM)	-5.26	-0.36	-4.81	0.11
Net Return (KS)	-1.09	-0.68	-1.61	-0.18

Table 8: Relative Strength Portfolios: Net Returns

This table presents returns net of cost in the post-formation week. Portfolios are formed every week. The sorts are based on returns, turnover, and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Return breakpoints are determined by the median return for both positive and negative lagged returns. Breakpoints for turnover and illiquidity are based on quartile breakpoints. Weights on stock i in week t are based on

$$w_{pit} = \frac{R_{it-1}T_{it-1}L_{it-1}}{\sum_{i=1}^{N_p} R_{it-1}T_{it-1}L_{it-1}},$$

$$p = WHH, WHL, WLH, WLL, LHH, LHL, LLH, LLL,$$

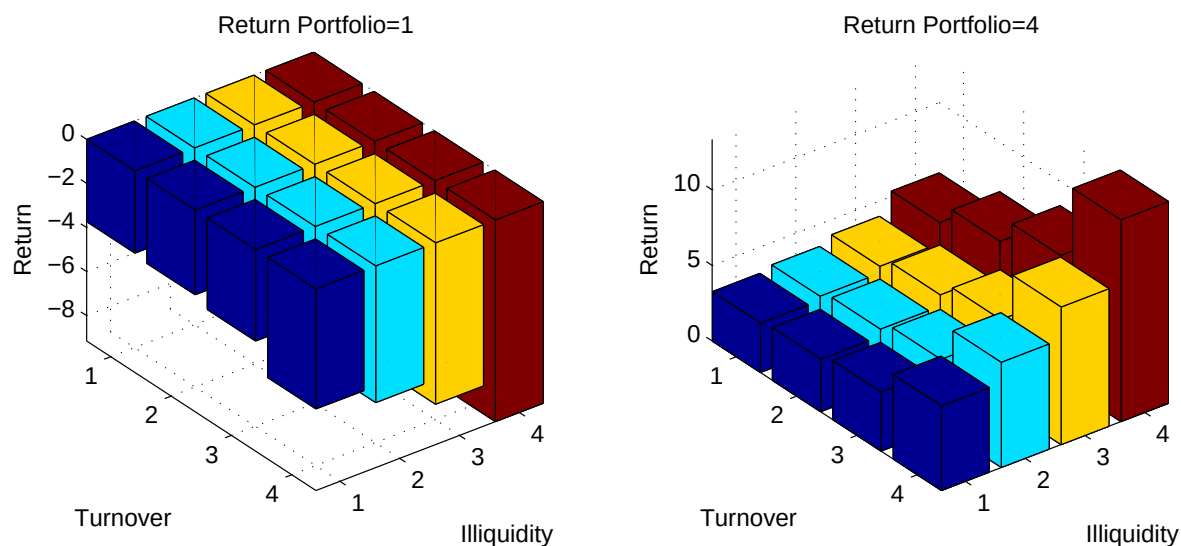
where T_{it-1} (L_{it-1}) represents the turnover (illiquidity) of stock i in week $t-1$ and N_p is the number of stocks in each of the eight portfolios formed by the three-way sort between winners (W) and losers (L) and high (H) and low (L) turnover and illiquidity. The results are presented only for the weighting criteria that utilizes return, turnover, and illiquidity. Portfolio returns are in percent per week and are based on skip-day methodology in which the return on the last day of the week is not used in computations. The returns are presented for a long position in return portfolio 1 (loser), a long position in return portfolio 4 (winner), and a combined long position in return portfolio 1 and a short position in return portfolio 4. Net returns are calculated by subtracting Keim and Madhavan (1997) and Korajczyk and Sadka (2004) (with a starting dollar amount of 10,000 dollars) cost estimates. These are indicated by the mnemonics ‘KM’ and ‘KS’, respectively. The sample period is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for all days of the week.

Turnover Portfolio	H	H	L	L
Illiquidity Portfolio	H	L	H	L
Return Portfolio=L (Loser)				
Raw Return	1.83	0.33	0.94	0.24
Net Return (KM)	-1.77	-0.39	-2.27	-0.02
Net Return (KS)	0.24	-0.34	-0.29	-0.24
Return Portfolio=W (Winner)				
Raw Return	-0.85	-0.07	0.15	-0.04
Net Return (KM)	-4.42	-0.78	-3.02	-0.26
Net Return (KS)	-2.15	-0.66	-0.80	-0.49
Return Portfolio=L-W (Loser-Winner)				
Raw Return	2.69	0.40	0.78	0.28
Net Return (KM)	-4.48	-1.03	-5.59	-0.21
Net Return (KS)	-0.20	-0.85	-1.39	-0.65

Figure 1: Three-Way Sorted Portfolios: Pre- and Post-Formation Returns

This figure shows portfolio returns in the pre- and post-formation week. Equal-weighted portfolios are formed every week. The sorts are based on returns, turnover, and illiquidity. Turnover is measured as the ratio of number shares traded to number of shares outstanding. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume). Return breakpoints are determined by the median return for both positive and negative lagged returns. Breakpoints for turnover and illiquidity are based on quartile breakpoints. Return portfolio 1 is the extreme loser portfolio and return portfolio 4 is extreme winner portfolio. Turnover (illiquidity) portfolio 1 has the lowest turnover (illiquidity), and portfolio 4 has the highest turnover (illiquidity). Portfolio returns are in percent per week and are based on skip-day methodology in which the return on the last day of the week is not used in computations. Stars indicate significance of means at 99%, 95%, and 90% level. The sample period is 1962 to 2002 and the portfolios include all NYSE/AMEX stocks that have data for all days of the week.

Panel A: Pre-Formation Returns



Panel B: Post-Formation Returns

